

Systems Biology: Mathematics for Biologists



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Last time: Differential equations of one variable

- Differential equations: $\frac{dx}{dt} = \dots$
- Solution: $x(t) = \dots$
- Often the solution is unknown.
- Qualitative analysis can tell us long-term behaviour.
- A **phase portrait** tells us where x increases, decreases and does not change.
- No change $\Leftrightarrow \frac{dx}{dt} = 0 \Leftrightarrow$ **equilibrium**.
- Equilibria can be stable or unstable.
- Equilibria can move when parameters move.
- Equilibria can appear or disappear, this is called a **bifurcation**.

Introduction to 2D systems

Contents

- Systems of two differential equations
 - Two co-dependent variables.
 - Linear and non-linear systems.
 - Generic solution of linear systems.
 - What do eigenvalues say?
- Qualitative analysis of system dynamics
 - 2D Phase portraits.
 - Finding equilibria.
 - Vectorfields.
 - Null-clines.
- General plan of analysis.
- Parameters and bifurcations in 2D.

One differential equation

$$\frac{dx}{dt} = f(x)$$

Note that there is just one variable, and that the function $f(x)$ depends only on x .

Thus the rate of change of x depends only on x itself.

Systems of two differential equations

$$\begin{cases} \frac{dx}{dt} = f(x, y) \\ \frac{dy}{dt} = g(x, y) \end{cases}$$

Note that there are two variables, and both the functions $f(x, y)$ and $g(x, y)$ depend on x and y .

Thus the rate of change of x and y depends on themselves and on each other.

Two example systems

Linear system — decaying and converting chemicals:

$$\begin{cases} \frac{dx}{dt} = ax + by \\ \frac{dy}{dt} = cx + dy \end{cases},$$

with $a = -2$, $b = 1$, $c = 1$, $d = -2$

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Non-linear system — predator-prey system:

$$\begin{cases} \frac{dx}{dt} = rx(1 - \frac{x}{K}) - bxy \\ \frac{dy}{dt} = cxy - dy \end{cases},$$

with $r = 3$, $K = 1$, $b = 1.5$, $c = 0.5$, $d = 0.25$

Solution of linear systems

For 1D linear systems

$$\frac{dx}{dt} = kx$$

the solution is given by

$$x(t) = Ae^{kt} \quad \text{with} \quad x(0) = A$$

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$$\begin{cases} \frac{dx}{dt} = ax + by \\ \frac{dy}{dt} = cx + dy \end{cases}$$

the solution is given by

$$\begin{aligned} x(t) &= C_1 v_{1x} e^{\lambda_1 t} + C_2 v_{2x} e^{\lambda_2 t} \\ y(t) &= C_1 v_{1y} e^{\lambda_1 t} + C_2 v_{2y} e^{\lambda_2 t} \end{aligned}$$

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with $x(0) = C_1 v_{1x} + C_2 v_{2x}$ and $y(0) = C_1 v_{1y} + C_2 v_{2y}$

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Finding a solution

Example find the solution of the following system:

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- Finding eigenvalues of $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$: $\det \begin{pmatrix} a - \lambda & b \\ c & d - \lambda \end{pmatrix} = 0$
- Finding eigenvectors:
 $\mathbf{v}_{1,2} = \begin{pmatrix} -b \\ a - \lambda_{1,2} \end{pmatrix}$ or $\mathbf{v}_{1,2} = \begin{pmatrix} -d + \lambda_{1,2} \\ c \end{pmatrix}$

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General solution:

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Finding a solution

Example find the solution of the following system:

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Note that, just as for 1D systems, is not possible to find an analytical solution for **non-linear** 2D systems.

Trajectories

Given the general solution:

$$x(t) = C_1 e^{-1t} - C_2 e^{-3t}$$

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and initial values $x(0) = 1$ and $y(0) = 2$

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$$x(t) = 1\frac{1}{2}e^{-1t} - \frac{1}{2}e^{-3t}$$

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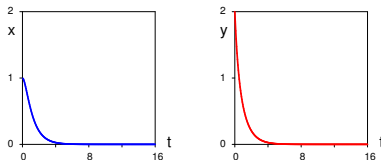
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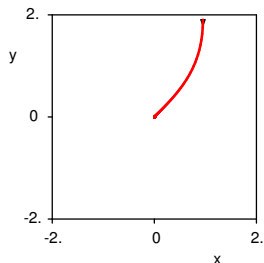
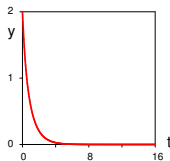
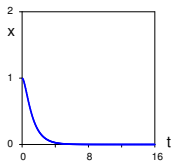
What dynamics do we expect x and y to follow?



Both eigenvalues are negative $\rightarrow$ decay to 0

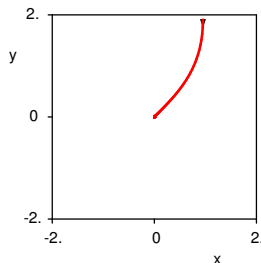
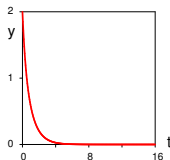
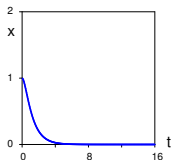
Phase portrait

We need a 2D phase portrait to depict the x and y dynamics together:



Phase portrait

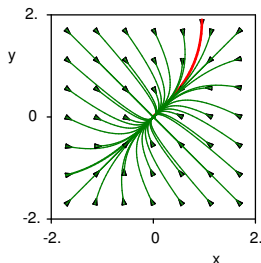
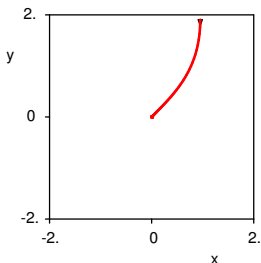
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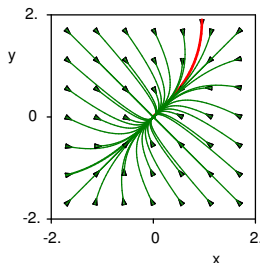
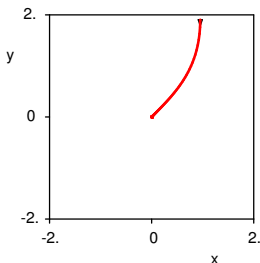
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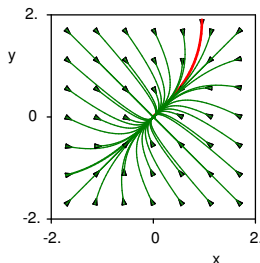
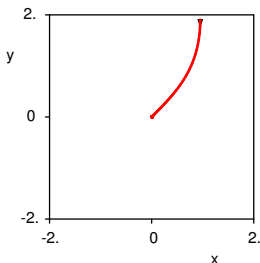


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Not so easy to draw properly without a computing device.

But general behaviour, convergence to $(0,0)$ follows from $\lambda_1 < 0, \lambda_2 < 0$.

Implies that $(0,0)$ is stable equilibrium.

Equilibria

For a 2D system, variables x and y are co-dependent:

$$\begin{cases} \frac{dx}{dt} = f(x, y) \\ \frac{dy}{dt} = g(x, y) \end{cases}$$

Therefore (x^*, y^*) is an **equilibrium** if *both*
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 $\frac{dx}{dt} = f(x^*, y^*) = 0$ **and** $\frac{dy}{dt} = g(x^*, y^*) = 0$

If only $\frac{dx}{dt} = f(x^*, y^*) = 0$ **or** $\frac{dy}{dt} = g(x^*, y^*) = 0$,
one variable changes, also causing the other variable to change.

Equilibria

Example: Find equilibria of the following two systems:

$$\begin{cases} \frac{dx}{dt} = -2x + y \\ \frac{dy}{dt} = x - 2y \end{cases}$$

$$\begin{cases} \frac{dx}{dt} = 3x(1-x) - 1.5xy \\ \frac{dy}{dt} = 0.5xy - 0.25y \end{cases}$$

Vectorfield

Qualitative phase portrait: only look at derivatives, depict with arrows.

$$\text{If } \frac{dx}{dt} = f(x, y) > 0$$

$$\begin{cases} \frac{dx}{dt} = f(x, y) \\ \frac{dy}{dt} = g(x, y) \end{cases}$$

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If $\frac{dx}{dt} = f(x, y) > 0$, x increases: $\rightarrow$

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If $\frac{dx}{dt} = f(x, y) < 0$, x decreases: $\leftarrow$

If $\frac{dy}{dt} = g(x, y) > 0$, y increases: $\uparrow$

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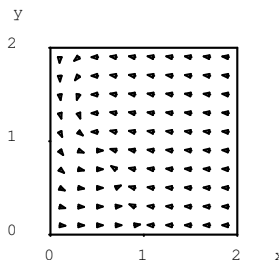
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Fill in x, y values to get $\vec{v} = (f(x, y), g(x, y))$.

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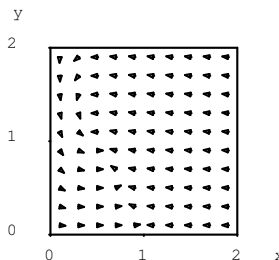
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Hard to see what long term dynamics will be!

Null-clines

Vectorfield: a lot of work, and still unclear.

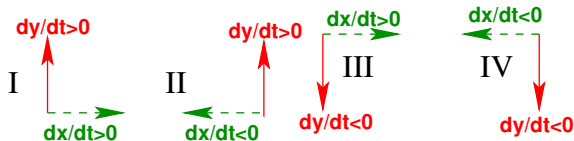
Can we do this in a smarter, more insightful way?

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Only 4 qualitatively different vectors are possible:

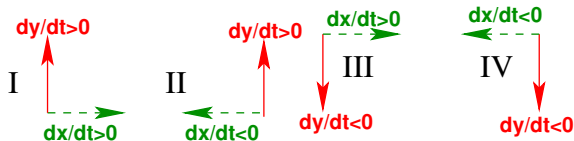


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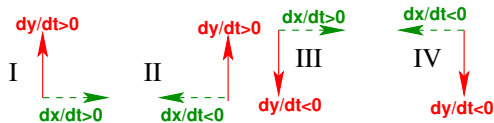
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Smarter approach:

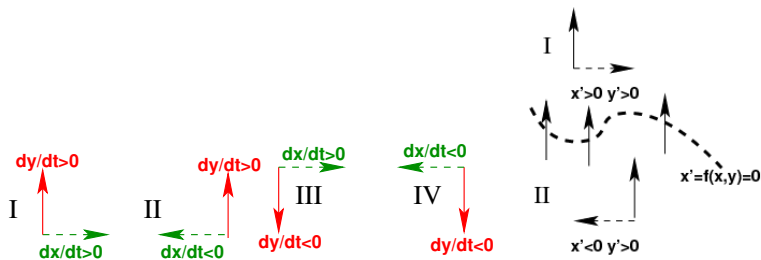
- 1 Divide vectorfield in different regions by finding boundaries.
- 2 Assign all vectorfield regions to one of these 4 categories.

Null-clines



From I to II, and from III to IV, the direction of $\frac{dx}{dt}$ changes.

Null-clines

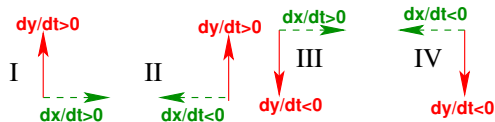


From I to II, and from III to IV, the direction of $\frac{dx}{dt}$ changes.

The **x null-clines** are the boundary lines at which $\frac{dx}{dt} = 0$ and the horizontal part of the vectorfield changes direction.

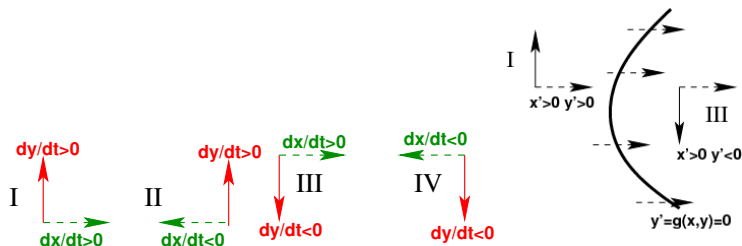
At an x null-cline, the vectorfield has only a vertical component.

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Null clines



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The **y null-clines** are the boundary lines at which $\frac{dy}{dt} = 0$ and the vertical part of the vectorfield changes direction.

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Combining null-clines and vectorfield

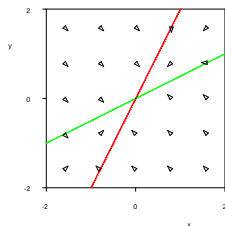
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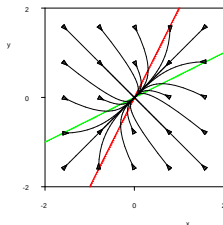
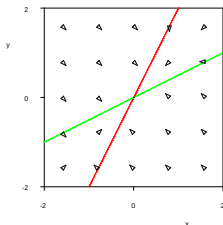
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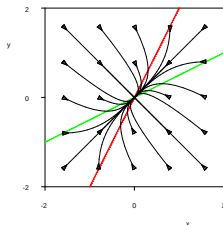
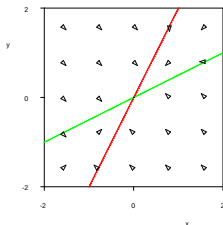
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Together, null-clines and vectorfield give more information!

Null-clines and equilibria

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If x and y null-clines intersect,
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If x and y null-clines intersect,
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Note that intersections of two nullclines *of the same type*
are no equilibria!

Global plan

General plan of qualitative analysis:

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 - Flip the horizontal component when crossing x null-cline.
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 - Flip the vertical component when crossing y null-cline.
- ④ Show the direction of the vectorfield on the null-clines.

Global plan

General plan of qualitative analysis:

$$\begin{cases} \frac{dx}{dt} = f(x, y) \\ \frac{dy}{dt} = g(x, y) \end{cases}$$

- 1 Draw $\frac{dx}{dt} = f(x, y) = 0$ and $\frac{dy}{dt} = g(x, y) = 0$ null-clines.
- 2 Choose a region of the x, y plane and find x and y direction vectors $\vec{v} = (f(x, y), g(x, y))$.
- 3 Find the vectorfield in the adjacent regions using:
 - Flip the horizontal component when crossing x null-cline.
 - Flip the vertical component when crossing y null-cline.
- 4 Show the direction of the vectorfield on the null-clines.
- 5 Try determine the dynamics around an equilibrium.

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- ⑥ Try to derive the global dynamics of the system.

Example

Example: Draw a vectorfield of the following system:

$$\begin{cases} \frac{dx}{dt} = 3x(1-x) - 1.5xy \\ \frac{dy}{dt} = 0.5xy - 0.25y \end{cases}$$

Parameters and bifurcations

Consider the following system:

$$\begin{cases} \frac{dx}{dt} = -ax + by \\ \frac{dy}{dt} = c \frac{x^2}{x^2 + d^2} - ey \end{cases}$$

a , b , c , d and e are all parameters.

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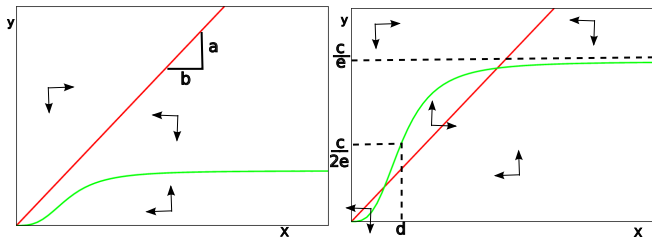
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Note that x and y both cause their own decrease.

Furthermore they both cause each others increase.

Parameters and bifurcations

Depending on the value of a , one or three equilibria:



So a determines if there is a non-trivial stable equilibrium.

In 2D systems as in 1D systems, a **bifurcation** is a **qualitative change** in the equilibria of a system.

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